



Linking modelling and empirical data to assess recreation services provided by coastal habitats: The case of NW Portugal

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ABSTRACT

In recent years, NW Portugal, particularly the city of Porto, has increasingly attracted tourists, thus emphasising the economic potential of the sector. This region is also of high ecological relevance, hosting a Marine Protected Area and several estuarine and coastal protected areas classified by the EU Natura 2000 programme. Therefore, there is a need to combine human activities and deliver societal benefits without compromising the ecosystem health and the maintenance of ecosystem services (ES). Here we give a novel local/regional assessment of the recreation and tourism magnitude in relation to ecosystem services using NW Portugal as a case study, including major estuarine habitats in the Douro estuary. We combined InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs) model predictions with available tourism data to interrogate, analyse and ensure that the Tourism Recreation model produced outputs maps of Photo-User-Day estimates. As a proxy for tourist visits, we used the geotagged photographs posted to the website Flickr™. The area being modelled was decided based on the location of natural habitats with a high attractiveness to visitors. We applied the Recreation model to different scales to compare the Photo-User-Day estimates over the local and regional scales of the study area to give a major focus for recreation associated with the Porto city and with the Douro estuary. This work shows the high value of the need for studies where ES are assessed at a local/regional scale and not based on extrapolations from general ES estimates, under the incorrect assumption that habitats worldwide are similar.

1. Introduction

Ecosystem services (ES) are produced by ecosystem processes and then societal benefits can be obtained from ES following the input of human complementary assets (Atkins et al., 2011; Elliott et al., 2017). Coasts and estuaries provide a great variety of ES, both material (e.g. food, water, energy) or non-material (e.g. recreation, cultural heritage) (Costanza et al., 1997; MA, 2005a; Barbier et al., 2011). The assessment and valuation of coastal and marine ES is vital, not only for promoting a better decision-making and sustainable development but also for knowing where to take actions to further guide policy-makers and stakeholders to better support decisions in the marine sectors (Liquete et al., 2016; Burdon et al., 2017). The Millennium Ecosystem Assessment (2005a,b) has encouraged policy-makers to consider the human dimension as an integrating component of ecosystems and thus challenging researchers to seek scientifically sound methodologies to be used to merge these aspects. Cultural ecosystem services result from the interaction between cultural practices and the natural environment (Fish et al., 2016). Such ES are included in all classifications (MA,

2005b; CICES (Haines-Young and Potschin, 2013) and are considered as the ‘ecosystem’ contribution to the non-material benefits as individual and collective human wellbeing, derived from human-ecosystem interactions (Chan et al., 2011; Breslow et al., 2016).

Coasts and estuarine ecosystems are important in delivering ES and societal goods and benefits (Costanza et al., 1997; Barbier et al., 2011), where large numbers often choose these ecosystems for their recreational activities (e.g. beaches, water sports, boat trips). Marine and coastal ecosystems are under threat and scientific consensus suggests and requires new holistic and adaptive environmental management approaches (Elliott and Whitfield, 2011; Elliott et al., 2017), focusing on ecosystems and their human, ecological, and physical components (Atkins et al., 2011; Barbier et al., 2011). The increasing world population and its concentration in coastal areas (Hugo, 2011) poses, even more, pressures on these ecosystems (Lotze et al., 2006; Halpern et al., 2008; Elliott et al., 2014, 2017), highlighting the urgent need to combine human activities with the delivery of ES and the maintenance of ecosystem health (Tett et al., 2013). These pressures, coupled with the importance that stakeholders and the public give to the non-material

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values derived by these ecosystems (Tielbörger et al., 2010; Milcu et al., 2013), emphasises the need for research in selecting the best approaches when valuing the ecosystems.

In recent decades, there has been a growing tendency of studies focusing on marine ES (e.g. Hernández-Morcillo et al. 2013; Liqueste et al., 2013; Lopes and Videira, 2013; Wakita et al. 2014; Martin et al., 2016), although marine cultural ecosystem services (MCES) are still studied less than other aspects (Schaich et al., 2010; Rodrigues, 2015). Furthermore, there is a disparity between terrestrial cultural ES and coastal and marine cultural ES (Liqueste et al., 2013; Lopes and Videira, 2013; Martin et al., 2016) with data or methods to assess MCES still further behind those for terrestrial ecosystems (Costanza, 1999; Markandya et al., 2008; Beaudoin and Pendleton, 2012; Barbier, 2012; Liqueste et al., 2013). Many factors potentially contribute to this such as the difficulty in evaluating the non-material ecosystems good and benefits (Kumar and Kumar, 2008; Chan et al., 2012; Tengberg et al., 2012; Milcu et al., 2013). The difficulty of translating aesthetics, feelings and other non-tangible aspects of nature into quantifiable values (Tengberg et al., 2012), still poses a great difficulty for researchers, even when a multidisciplinary approach is used (Sessions et al., 2016).

The cultural ecosystem service more amenable for evaluation and assessment is that for tourism and recreation, which in the ES context refers to a wide array of public, local, nature-based activities and outdoor recreational activities with an important role in human wellbeing and health (Maes et al., 2011). In fact, tourism and recreation as ES are more amenable to valuation, are easier to quantify, and often generate high revenues to the global economy (Costanza et al., 1997; de Groot et al., 2012). This can then help to explain the reason for cultural ES studies being mostly focused on the economic valuation of nature-based recreation and tourism (Milcu et al., 2013).

Marine and estuarine ES is an emergent research field, presenting new challenges and opportunities for science and the application of the ecosystem services framework (Barbier et al., 2011; Guerry et al., 2012; Burdon et al., 2017). Despite this, ES studies need to evolve from extrapolations from general ES estimates, under the incorrect assumption that habitats worldwide are similar, to studies where the ES are quantified at a local/regional scale. Hence, this study aimed to assess the recreation and tourism ecosystem services of Northwest of Portugal, by combining modelling and empirical data. Specifically, two different approaches were used to assess recreation and tourism nature-based services: i) using the InVEST Recreation model, and ii) analyzing socio-economic tourism indicator data from official sources. This study represents the first local/regional assessment of the recreation and tourism provided by coastal and estuarine habitats of NW Portugal and will improve the accuracy of the tourism and recreation global/large scale assessments.

2. Material and methods

2.1. Methodological approach

To assess the recreation services provided by NW coastal habitats of Portugal we combined two different methodologies. First, we used the Recreation InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs) model methodology (Sharp et al., 2016) to calculate the number of photos in Flickr per grid cell on the sub-regions, to assess the number of visitors in the region and to calculate coverage of each predictor for latter analysis of visitors preferences. Then we used official tourism socio-economic indicators to compare the temporal trends in visitor numbers in the study area given by the InVEST model. Given the growing trend of tourism and recreation activities in NW Portugal over the past decade (INE, 2016), we focused this study on the period 2005–2014. The study area (Fig. 1) was divided into 4 sub-regions of different scales: *sub-region 1*- considering the entire NW of Portugal, comprising the districts of Porto, Braga and Viana do Castelo; *sub-region 2*- focusing only the civil parishes bounding coastal and the five

estuarine areas of the NW of Portugal; *sub-region 3* - comprising the Porto district, and *sub-region 4* - defined by the civil parishes that bound the entire Douro estuary. Sub-regions 2 and 4 were designed to include the tourism and recreation services directly provided by the coastal and estuarine ecosystems of each of the main areas, namely sub-regions 1 and 3. In particular, the NW of Portugal (*sub-region 1*) has several other natural landscapes providing tourism and recreation apart from coastal and estuarine ecosystems. Therefore, in order to assess visits to coastal and estuarine ecosystems, we used a narrow sub-region that only included the coastline and the 5 major estuaries of the NW Portugal/*sub-region 2*). A similar approach was applied to the Porto city (*sub-region 3*), that together with the Douro estuary (*sub-region 4*) are the focus of most of the visit numbers according to official data (INE, 2016). A buffer of 5 Km from the shoreline into the Atlantic Ocean was also created to include photographs taken from boats.

2.2. Study area

The NW coast of Portugal (Fig. 1) has notably increased in tourism and recreation intensity over the past decade (INE, 2016), a result of the local initiatives to promote tourism mainly in the city of Porto. This area is considered the most developed of the north of Portugal, hosting the second largest city in Portugal and important urban and industrial areas. This area is characterised by a mosaic landscape of cities, villages, broad leaf and coniferous forests, varied agriculture and vineyards, and with variable elevation (CORINE Land Cover 2006; Caetano et al., 2009). The area also has very extensive protected areas under the Natura 2000 classified areas (including both the EU Habitats and the Birds Directives), and one marine protected area, the 18 Km long Parque Litoral Norte. Five main river estuaries are in this region: the Minho, Lima, Cávado, Ave and Douro. These estuaries are subject to human activities and the Douro has been considered as the most impacted system (Ramos et al., 2015). The coastal region is mainly dominated by sand beaches with patches of rock shores and cliffs (Taveira Pinto, 2004). The large local and international tourism attractions are the coastal beaches and natural places, together with the historical cities (e.g. Braga, Guimarães and Viana do Castelo), and the city of Porto, and the Douro River and the Alto Douro wine region (INE, 2016). These UNESCO World Heritage Sites attract increasing tourists annually (INE, 2016) associated with both natural landscapes and Port Wine Culture and Industry.

2.3. InVEST tourism and recreation model

InVEST is an open-source tool to map, quantify and value Ecosystem Services. It is flexible and consists of scientifically grounded computer-based models, focused on ecosystem services derived from the underlying biophysical processes, and provides outputs in both biophysical terms and monetary and non-monetary value (Guerry et al., 2012). The InVEST suite of models developed by the Natural Capital Project (www.naturalcapitalproject.org) has been used in several studies targeting ES (Kareiva et al., 2011; Chung et al., 2015; Sharp et al., 2016), and also has been used in reliable management and decision making, since these models provide spatially explicit and quantified outputs of ES, suitable for a broad understanding (Sharp et al., 2016).

The InVEST Recreation model (Sharp et al., 2016) was used to estimate the visit rate of the study area. This model uses geotagged photographs from the Flickr website (<https://www.flickr.com>) as a proxy indicator for the annual number of visitors at a given location (Wood et al., 2013; Keeler et al., 2015). The model estimates visit rates across a defined area (grid cells or polygons) in the form of photo-user-days (PUD, i.e. 1 PUD refers to one person in one unique place on one unique day of the year) (Sharp et al., 2016). The cells with no PUD were included in the further analysis.

The InVEST regression analysis included in the Recreation model was used to estimate the importance of potential landscape features

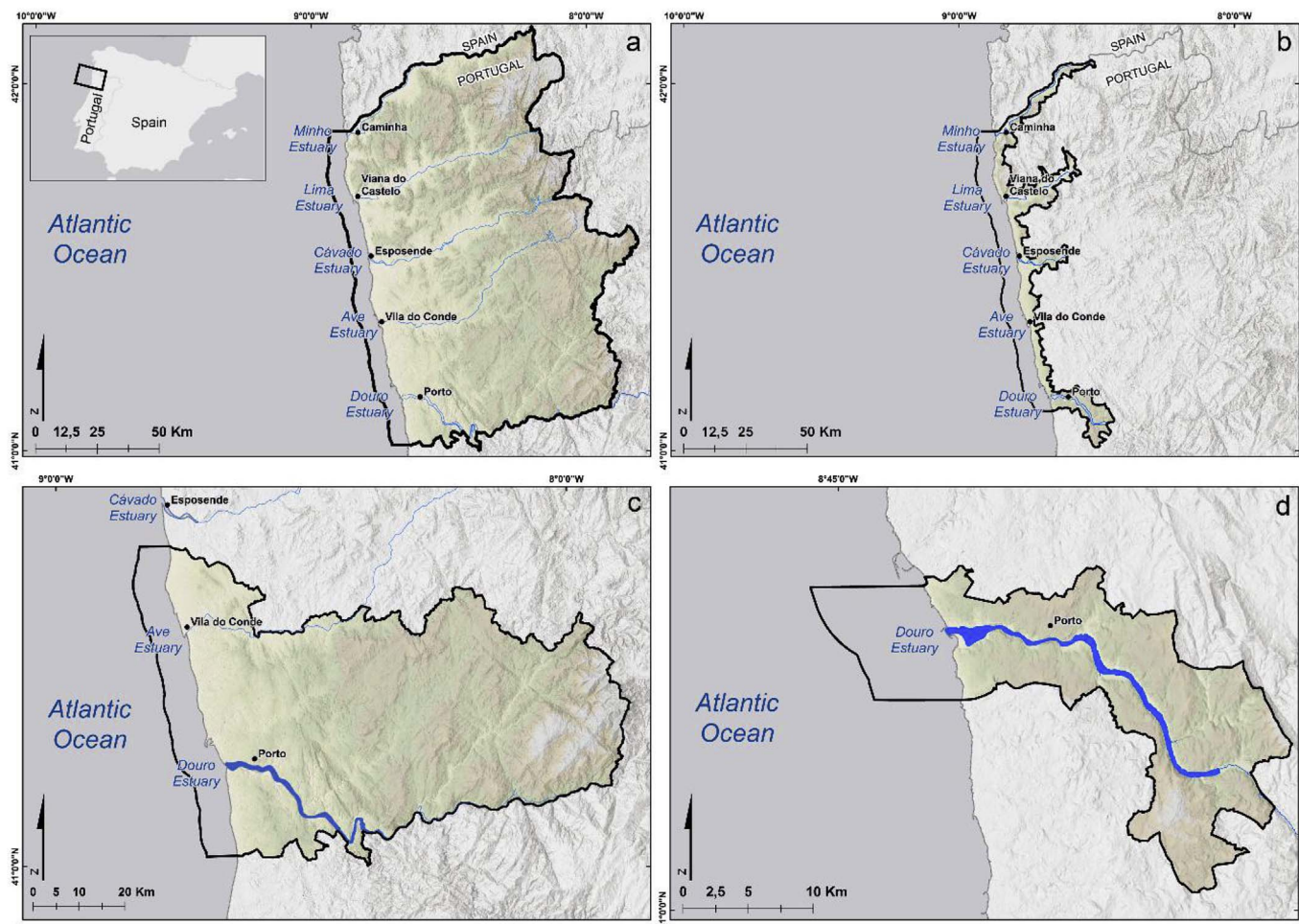


Fig. 1. Location of the sub-regions of the study, a – sub-region 1, b – sub-region 2, c – sub-region 3, d – sub-region 4. Data Credits: ICNF, EU, USGS, NGA, NASA, CGIAR.

Table 1
List of landscape features and respective data source used in the InVEST model. (Note: Variables in bold were excluded from the model).

Predictor	Description	Source
Land cover	Corine Land Cover 2012, using Artificial, Agricultural, Forest and Semi-Natural, Wetlands and Waterbodies categories	http://land.copernicus.eu/pan-european/corine-land-cover/clc-2012
Estuaries	Estuaries boundaries	http://snirh.apambiente.pt/
Beaches	Beaches, digitised	http://snirh.apambiente.pt/
Protected areas	National and Listed protected areas	http://www.icnf.pt/portal
Natural areas	Natural or Semi-Natural areas not included in other sources (e.g. green urban areas)	http://www.protectedplanet.net
Viewsheds	Viewsheds from the city of Porto	www.openstreetmap.org
Port Wine Cellars	Port Wine Cellars Addresses	http://www.visitporto.travel
Accessibility (roads network)	Network of primary, secondary and highways	http://www.aevp.pt/ www.openstreetmap.org

(e.g. natural landscapes, accessibility by the road network, protected areas) to explain the spatial distribution of visitors in each of the 4 sub-regions of the study area. This analysis used a log-transformed PUD value per cell +1 (dependent variable), and then performs a simple linear regression to infer the effect of each landscape attribute (independent variable) on those visit rates (Sharp et al., 2016). The model predictors selected in this study (Table 1) were based on the criterion of being associated with natural coastal and estuarine ecosystems (e.g. protected areas), but also for their relevancy for the study area tourism (e.g. Port wine cellars). Each of the four sub-regions was divided into grid cells of 1 km², except sub-region 4 that was divided into 500 m² grid cells. The correlation between predictors was assessed using a Pearson correlation coefficient (Fig. 2), and the Variance Inflation Index (VIF) was also calculated for each of the predictors in the linear

regression for a further analysis of possible multicollinearity between predictors. As the correlation analysis showed a major correlation between ‘Viewsheds’ and ‘Port Wine Cellars’, we choose to exclude this predictor from further analysis. A significance level of $p = .05$ was used for all the analysis.

2.4. Socio-economic tourism indicators

Official data for tourism of the NW coast of Portugal was also gathered from the Portuguese Institute of Statistics (INE, 2016) to complement information given by the InVEST Recreation model. Available data were limited, with several socio-economic indicators only being available at a national scale, or for a limited period different from 2005–2014. Therefore, for our study area and for the period

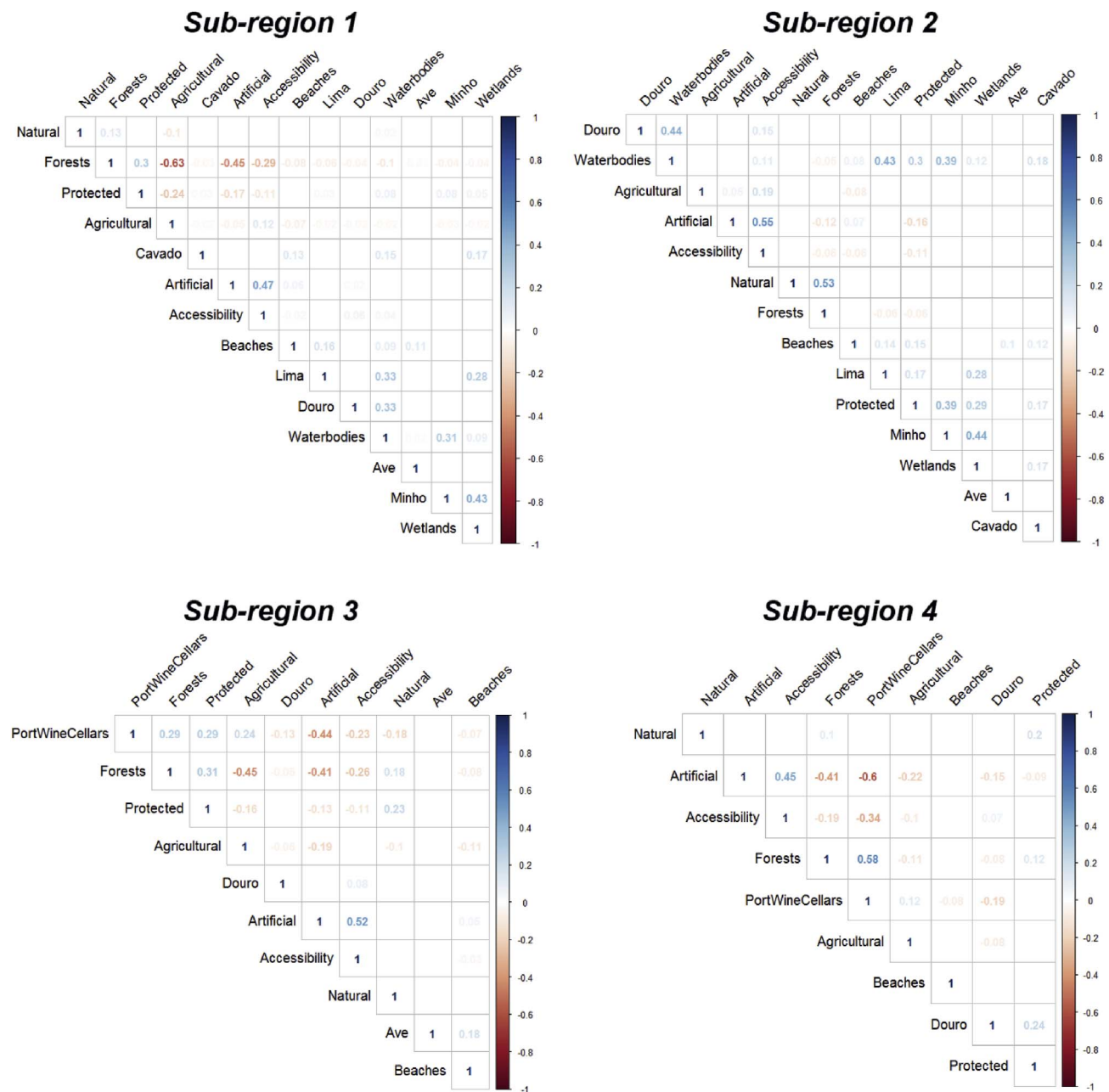


Fig. 2. Correlation matrices between the predictors used in each sub-region. Strong colours indicate stronger correlation. The blank squares indicate non-significant correlations. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2005–2014, only two indicators had annual available data and so were selected: (1) total number of guests (INE, 2016) and (2) number of visitors to Port Wine Cellars (AEVP – Port wine Companies Association). These socio-economic tourism indicators were used to compare the annual trend of tourism assessed by the InVEST model results.

3. Results

3.1. Socio-economic tourism indicators

The matching of the socio-economic tourism indicators used (total number of guests) was only possible for sub-region 1 and 3. The number of visitors to Port Wine Cellars was only compared to sub-regions 3 and 4, given the specificity of this indicator to the sub-regions. For sub-region 1, the total number of Guests increased during the study period (Fig. 5), and especially after 2012. In contrast, the PUD number

increased until 2011, but reduced after 2011 (Fig. 5). Both the total number of guests and the total number of visits to Port Wine Cellars also increased over the period analyzed for sub-regions 3 and 4 (Fig. 6). For sub-region 3 the number of guests doubled in 2014, compared to 2005 (from 1 022 259 guests in 2005 to 2 200 896 guests in 2014). Port wine Cellar visitors fluctuated over the years, but there was an overall increase of almost 37% more visitors in 2014 compared to 2005. The change over time of both Number of Guests and the Port wine visitors did not agree with the temporal visit rates estimated by the InVEST model (Figs. 5 and 6). Socio-economic tourism indicators showed increasing tourism during the study period, with higher values registered in 2014, in contrast with the PUD estimates that showed decreasing PUD numbers after 2011 in sub-regions 3 and 4.

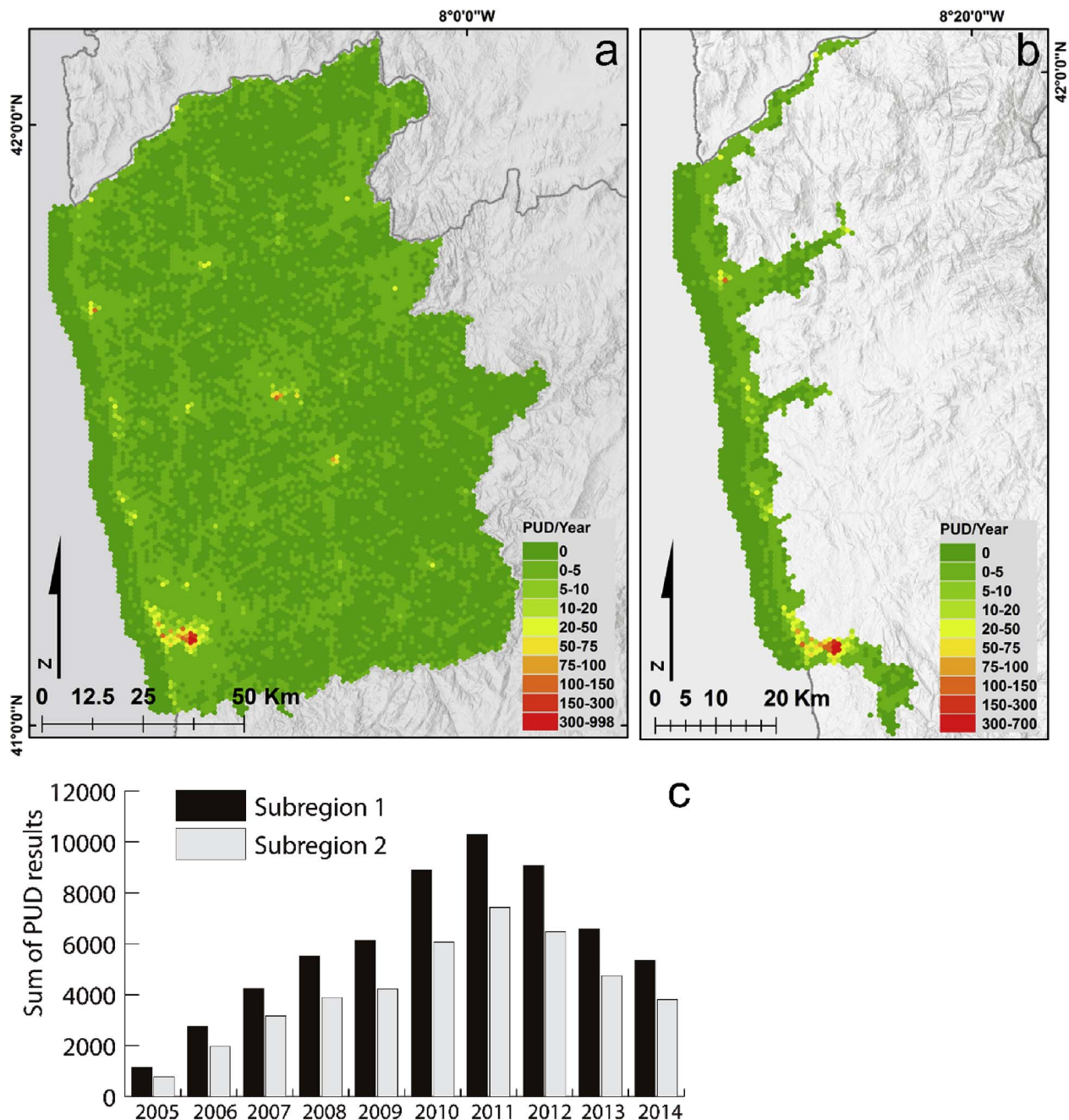


Fig. 3. InVEST model average PUD results for sub-region 1 and 2. (a) 10-year average PUD results for sub-region 1; (b) 10-year average PUD results for sub-region 2; (c) sum of the average PUD results for each year, in both sub-regions. Data Credits: ESRI. HERE. DeLorme. MapmyIndia. © OpenStreetMap.

3.2. InVEST model results

3.2.1. Sub-regions 1 and 2

The average visit rates from 2005–2014, estimated by the InVEST model for the sub-region 1, was 6008 PUD/year and 4257 PUD/year for the sub-region 2. The visit rate of sub-region 2 represented 71% of the total visit rate of the NW Portugal (sub-region 1). PUD maps of the sub-regions 1 and 2 (Fig. 3) showed that coastal cities of Porto, Matosinhos and Viana do Castelo had the highest visit rates, although there were some cells reaching high PUD values located in some inner cities, such as Braga, Guimarães and Ponte de Lima (Fig. 3). The highest cell value of the 10-year average PUD results was for Porto (highest PUD = 998) followed by Viana do Castelo (highest PUD = 111). Sub-region 1 had an average of 62% of zero PUD cells (no. of cells = 7226) (Fig. 3a), while in the sub-region 2 this ‘no visit rate’ only represented 49% of the

territory (no. of cells = 724) (Fig. 3b). The temporal pattern of the InVEST model visit rates results for the two sub-regions 1 and 2, shows that the annual PUD results increased from 2005 to 2011, and then decreased until 2014 (Fig. 3c). The PUD values and average annual guests between 2005 and 2011 showed a correlation of $r = 0.94$ ($p < .05$) ($r = 0.44$, $p = .20$, when analyzing 2005–2011).

The selected landscape attributes used in the model explained 40% ($R^2 = 0.40$) of the obtained visit rates for the sub-region 1, and 55% ($R^2 = 0.55$) for the sub-region 2 (Table 2). For sub-region 1, all landscape attributes were significantly correlated with visit rates, except for the *Minho and Lima estuaries* and *Wetlands* (Table 2). All the predictors were positively correlated with the visitor numbers, except the Lima River, *Forests and Semi-Natural Areas* and *Agricultural Areas*. Similarly, most landscape attributes used in the model for the sub-region 2 were significantly correlated with the visit rates, except for the *Minho and Lima*

Table 2

Multiple linear regression model results from InVEST Recreation model quantifying the relationship between visit rates (log-transformed PUD) and landscape features for all the sub-regions of NW Portugal. * $p < .05$; ** $p < .01$; *** $p < .001$; na – not applicable.

Predictors	Sub-region 1			Sub-region 2			Sub-region 3			Sub-region 4		
	Estimate	SE	VIF	Estimate	SE	VIF	Estimate	SE	VIF	Estimate	SE	VIF
(Intercept)	4.33E-02 **	1.36E-02	na	3.92E-02	2.54E-02	na	1.16E-01***	2.56E-02	na	6.03E-01***	8.45E-02	na
Artificial Areas	8.54E-03***	2.16E-04	2.006	1.55E-02***	6.95E-04	1.536	8.22E-03***	3.73E-04	2.05	5.76E-03***	8.54E-04	2.18
Accessibility (roads network)	7.96E-05***	4.36E-06	1.333	1.71E-04***	1.92E-05	1.563	1.17E-04***	8.19E-06	1.41	5.34E-04***	5.25E-05	1.29
Agricultural Areas	−7.53E-04***	1.59E-04	2.342	−3.44E-03***	5.54E-04	1.06	−8.94E-04**	3.02E-04	1.89	−2.56E-03**	9.16E-04	1.15
Beaches	6.47E-02***	2.12E-03	1.093	5.04E-02***	3.78E-03	1.108	8.07E-02***	3.54E-03	1.07	4.10E-02***	7.70E-03	1.01
Estuaries												
Minho estuary	1.23E-03	1.78E-03	1.528	3.15E-04	3.71E-03	2.217	na	na	na	na	na	na
Lima estuary	−1.01E-03	1.40E-03	1.396	−1.80E-04	2.82E-03	1.994	na	na	na	na	na	na
Cávado estuary	1.53E-02***	2.85E-03	1.121	1.48E-02**	5.11E-03	1.263	na	na	na	na	na	na
Ave estuary	1.58E-02***	4.12E-03	1.015	1.62E-02*	6.90E-03	1.016	3.40E-03	4.45E-03	1.04	na	na	na
Douro estuary	3.29E-02***	1.55E-03	1.179	2.75E-02***	3.08E-03	1.634	3.70E-02***	1.79E-03	1.04	9.65E-03***	1.53E-03	1.30
Forests and Semi-Natural Areas	−4.97E-04***	1.48E-04	2.965	−1.05E-03	6.40E-04	1.445	−8.17E-04**	2.94E-04	2.19	1.72E-03	8.79E-04	1.69
Natural Areas	1.03E-03***	2.36E-04	1.023	2.73E-03*	1.09E-03	1.416	7.99E-04	5.76E-04	1.21	4.59E-03	2.41E-03	1.08
Port Wine Cellars	na	na	na	na	na	na	−3.03E-06***	6.21E-07	1.70	−7.62E-05***	8.97E-06	2.27
Protected Areas	6.09E-04***	4.79E-05	1.131	9.16E-04*	4.33E-04	1.372	1.38E-03***	3.94E-04	1.29	6.20E-03	3.46E-03	1.14
Water Bodies	7.99E-03***	8.29E-04	1.615	7.51E-03**	2.62E-03	2.95	na	na	na	na	na	na
Wetlands	2.76E-03	1.92E-03	1.509	1.24E-03	3.47E-03	1.733	na	na	na	na	na	na
Residual Standard Error	0.35 on 11581 df			0.59 on 1450df			0.43 on 3778df			0.75 on 1019df		
Adjust R-Squared	0.4			0.55			0.48			0.46		

estuaries, Wetlands and Forest and Semi-Natural Areas (Table 2). The regression analysis showed that *Beaches* was the most important attribute to explain visit rates of both *sub-region 1* (coefficient = $6.47E^{-02}$) and *sub-region 2* (coefficient = $5.04E^{-02}$) (Table 2). The Rivers Douro, Ave and Cávado also showed high estimates in the regression analysis with the visit rates in both sub-regions 1 and 2.

3.2.2. Sub-region 3 and 4

For the sub-regions 3 and 4, the InVEST model indicated an average visit rate of 4187 PUD/year and 3159 PUD/year, respectively, between 2005 and 2014. *Sub-region 4* exhibited a visit rate that was 75% of the *sub-region 3*. The Douro estuary (*sub-region 4*) accounted for 52% of the overall visit rate of the entire NW of Portugal (*sub-region 1*). PUD maps showed that higher PUD values were observed at the Porto (highest PUD value of 1053), Matosinhos (highest PUD value of 60), Vila do Conde (highest PUD value of 32) and Póvoa de Varzim cities (highest PUD value of 17) (Fig. 4). The higher PUD values tended to be along the Douro estuary and in the coastal areas of these two sub-regions (Fig. 4). For an absence of visits, 2229 zero PUD value cells were observed in *sub-region 3*, corresponding to 59% of the total sub-region area (Fig. 4a). For the *sub-region 4*, only 584 zero PUD value cells were estimated by the model, representing 57% of the total sub-region area (Fig. 4b). There was also an increasing visitor rate from 2005 until 2011, but then it started to decrease until 2014 (Fig. 4c).

The selected landscape attributes used in the model explained 48% ($R^2 = 0.48$) of the obtained visit rates for the *sub-region 3* and 46% ($R^2 = 0.46$) for the *sub-region 4* (Table 2). Most selected attributes were significantly and positively correlated to the visit rate of the *sub-region 3* (Table 2), and *Beaches* was the most important attribute (coefficient = $8.07 E^{-02}$). For *sub-region 3*, our analysis showed negative estimates for *Agricultural Areas*, *Forest and Semi-Natural Areas* and *Port Wine Cellars*. A similar scenario was observed for the *sub-region 4*, with most selected landscape attributes being significantly and positively correlated with the visit rate estimates (Table 2). The regression analysis of the *sub-region 4* showed that *Beaches* was also the most important attribute (coefficient = $4.10 E^{-02}$) with *Port Wine Cellars* and again, *Agricultural Areas* being the only attributes negatively correlated with the visit rate of the *sub-region 4* (Table 2).

4. Discussion

4.1. Comparison of both approaches

In the past decade, tourism of the NW of Portugal and particularly, the Porto area has increased exponentially, reflecting the international awareness of Porto and Alto Douro as UNESCO World Heritage and Porto as the 2001 European Capital of Culture, prompted by the city investment in tourism promotion initiatives (Meijers et al., 2012). The InVEST Recreation model identified the estuarine and coastal ecosystems (*sub-region 2 and 4*) as major contributors for the recreation of the NW Portugal and Porto district areas, which accounted for > 70% of the total visit rates of the all sub-regions areas (sub-regions 1 and 3). Notably, the Douro estuary accounted for 75% of the visit rates of the Porto district and 52% of all the NW of Portugal, emphasising the importance that coastal and estuarine ecosystems have on the recreational potential and economic revenues of the region.

The two methodologies used here show two different patterns of recreation. The socio-economic indicators showed a clear increasing trend of visitors, while the InVEST Recreational model results showed a growing tourism activity in all the *sub-regions* until 2011, and a decline after that year. After 2011, the trend in visit numbers derived from Flickr photo uploads differed from the indicators of official data sources on tourism for our particular study area. The estimated decrease in the visit rate by InVEST differed from the continued growing tendency of visitors indicated by the official data. Despite the capacity of the model to capture the visitor increase until 2011, from that year onwards the model diverged from official socio-economic indicators. This may be related to the emergence of other social media platforms such as Facebook (2004) and Instagram (2010). Despite the differences between PUD values and total guests in our study area, the relationship between visitors to a region and photograph uploads to Flickr is well documented (Wood et al., 2013; Keeler et al., 2015; Sonter et al., 2016). However, it is noted that Flickr users and photograph uploads do not necessarily represent the entire visitor population of a region. This, of course could introduce bias in these assessments although other more conventional methodologies such as survey data are also not free of bias (Hanemann, 1994; Hollenstein and Purves, 2010; Kling et al., 2012; Wood et al., 2013; Keeler et al., 2015). For example, the non-responsive or hasty/bad responses to survey questions could also bias survey results, especially those focused on willingness-to-pay (WTP) values

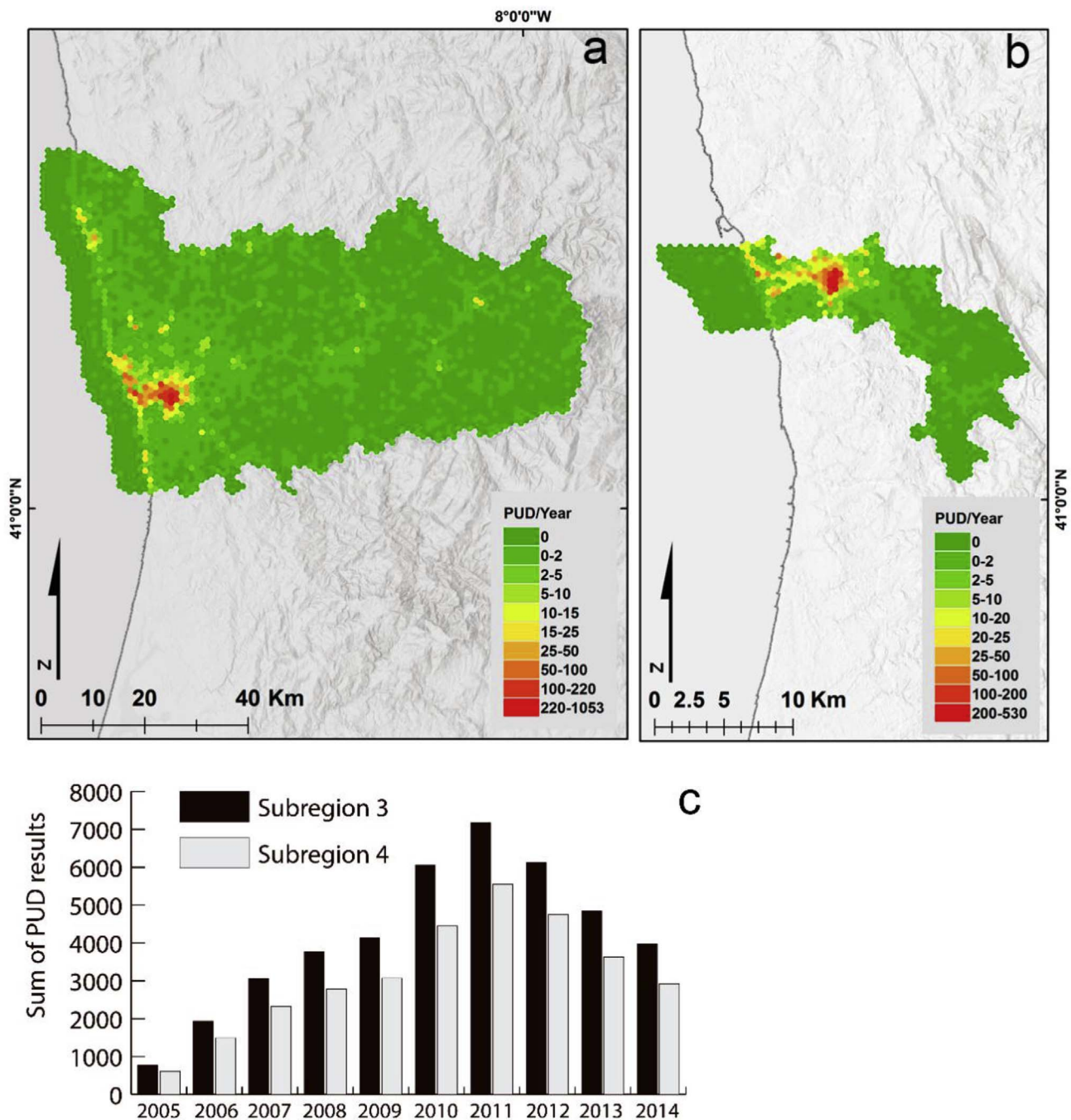


Fig. 4. InVEST model average PUD results for sub-region 3 and 4. (a) 10-year average PUD results for sub-region 3; (b) 10-year average PUD results for sub-region 4; (c) sum of the average PUD results for each year, in both sub-regions. Data Credits: ESRI. HERE. DeLorme. MapmyIndia. © OpenStreetMap.

(Rodrigues, 2015; Pinto et al., 2016).

4.2. Landscape attributes analysis

Coasts and estuaries, being the interface of seas, rivers and land ecosystems, provide innumerable recreational opportunities (Pendleton, 2009; Barbier et al., 2011). Estuaries and coastal areas, and especially beaches, are major landscape attributes in all four sub-regions models that influence societal choices for recreation. In particular, in our study beaches were the predictor most correlated with the visit rates (Table 2), followed by some of the main estuaries in the region, mainly the Douro, Ave and Cávado estuaries, highlighting their importance to the recreational activities in the region, as shown elsewhere (Hill and Courtney, 2006; Weyland and Laterra, 2014). People also tend to

choose nearby places for recreation, and accessibility plays a major role in their choice for recreation areas (Clark and Stanley, 1979; Maes et al., 2011; Parachini et al., 2014); this is shown by accessibility and artificial areas being important predictors of visit rates in our regressions. All the higher PUD value cells were located within or near cities, villages and roads throughout the region (Figs. 2 and 3). In being regarded as attractive sites for recreation (Bartczak et al., 2008), forests and semi-natural areas were negatively correlated with visitor numbers in our sub-region, except for sub-region 4. This might be due to the fact that the most important forests and natural parks of the NW Portugal are in the countryside, in more remote and inaccessible locations away from the coast. The remaining areas of forests are mainly for producing eucalyptus (*Eucalyptus globulus*) or maritime pine (*Pinus pinaster*) for historical paper production (ICNF, 2013), with the areas being not

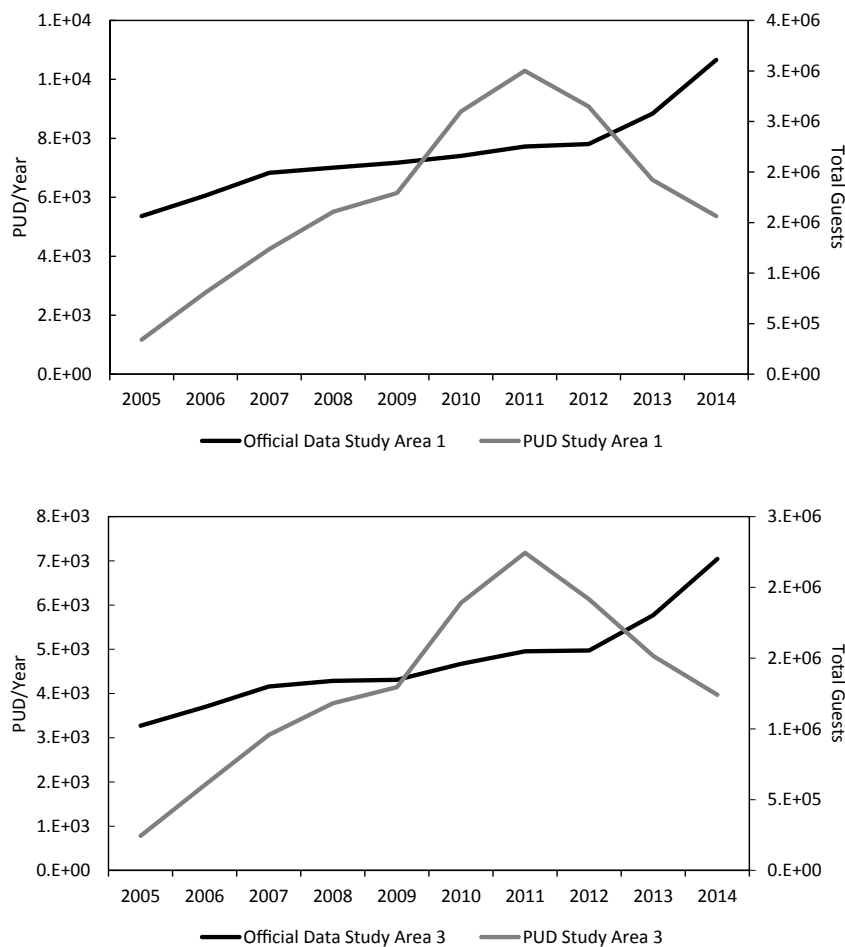


Fig. 5. Comparison of trends between InVEST Recreation model results and total number of guests for sub-region 1.

particularly attractive for recreational activities. The smaller area of sub-region 4 and its location centred on a main city, could have led to the positive relationship with visitor numbers, since city dwellers could prefer to spend their recreation time in green landscapes. This fact was also shown in the results of the predictor *Natural Areas*, representing the spatial distribution of cities green parks. This predictor showed positive estimations associated with the visitor numbers in all the four sub-regions analyzed. In addition, the predictor *Protected Areas* was positively correlated with visitor rates in all the sub-regions. It is of note that the visitor rate values within all protected areas (the sum of PUD values found in cells within protected areas reached a value of 1290 PUD in the 10-year average). This was almost wholly due to the estuarine and coastal protected areas (914 PUD), reinforcing the importance of these ecosystems for the tourism and recreation of the area.

The city of Porto is known for its architectural heritage but mainly by its intimate links to the Douro estuary and the Port Wine Industry and Culture of the Alto Douro vineyards (World Heritage). However, our results did not reinforce this feature, since *Port Wine Cellars* were shown not to be a significant predictor of the visit rates, in both sub-regions 3 and 4. This might have been a methodological issue associated with the small size of the cells used in InVEST model of the sub-region 4, and to the fact that most Port wine cellar pictures are taken from other different locations than the cellars, namely from the Porto city side of the Douro estuary.

4.3. Methodological constraints

The methodological approach selected for this study combined modelling and empirical data to enable a local assessment of tourism and recreation provision in coastal and estuarine ecosystems. The

InVEST model used might have underestimated the visitor rates after 2011, due to a possible model constraint associated with only using Flickr uploads as a data source. This constraint was identified by combining the model outputs with official tourism data, thus emphasising the importance of using different methodologies in ES assessments. InVEST model tools are an important aid for the first assessment of ecosystem services, but the additional linkage with other traditional sources of data, such as surveys, or non-traditional sources such as social media data and other indirect sources of data, becomes a powerful tool in decision-making at local scales. In general, the use of social media information or other sources of the so-called 'big data' can be a powerful tool in cultural ES studies. This can assist in bypassing the lack of data, and understand people's choices in recreation and connection to Nature, as is necessary to prioritise areas of intervention and define accurate and sustainable management and exploitation plans. Furthermore, some limitations on the model itself should be mentioned. Sharp et al. (2016) indicate that the model does not presuppose that any of the predictors chosen have any effect on visits, but that the model just estimates the extent of each predictor effect based on its spatial correspondence with the previous calculated visitors count in each of the grid cells. However, given our knowledge of the study region, the regression outputs from the InVEST model do not deviate from that expected. Despite some of the limitations, this approach provides a low-cost first assessment and indicator for mapping the cultural ecosystem benefits provision (Tenerelli et al., 2016).

4.4. Decision-making and management implications

The present study indicated the importance of coastal and estuarine ecosystems for delivering nature-associated recreational opportunities

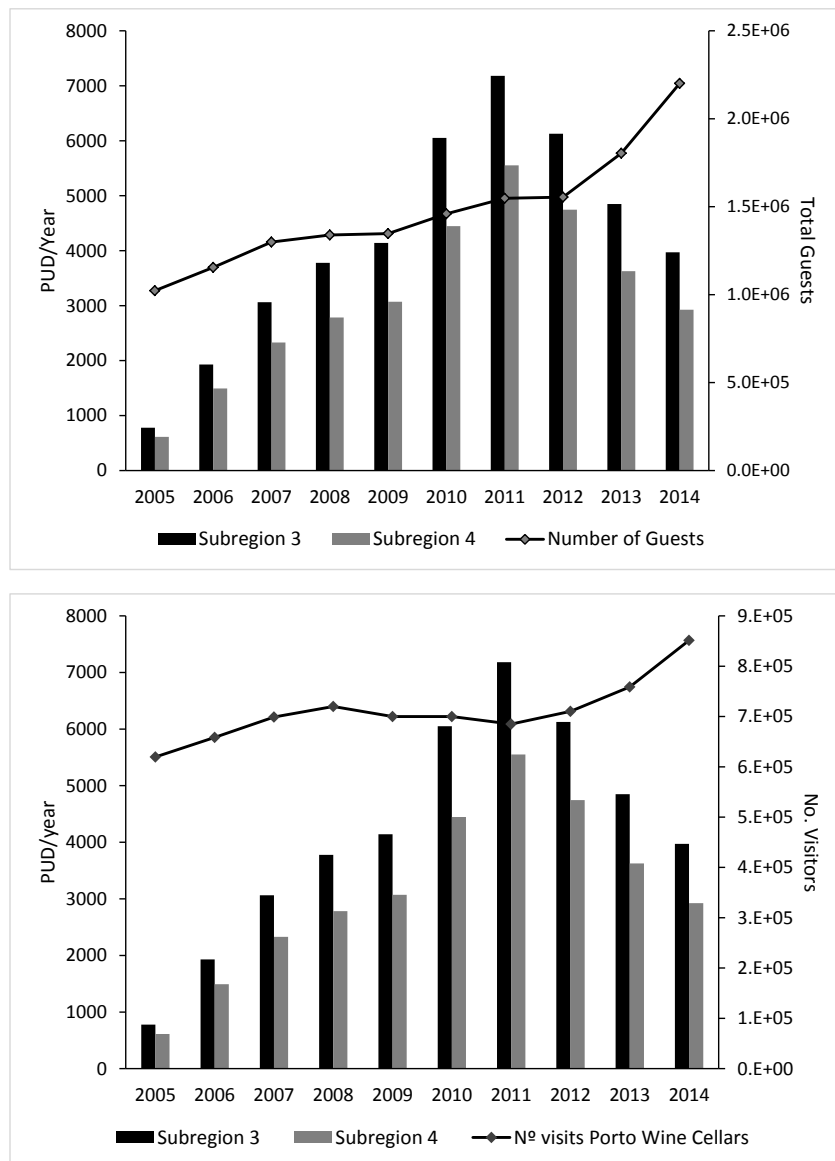


Fig. 6. Comparison of trends between InVEST Recreation model results and socio-economic indicators used for sub-regions 3 and 4.

(Rees et al., 2010; Ghermandi et al., 2011; Ahtiainen et al., 2013; Fletcher et al., 2014). Estuaries and coasts encompass pressures that risk their integrity and conservation (Elliott and Whitfield, 2011) but they are an important focus of recreational activities. Despite being one of the most important activities contributing to the national Gross Domestic Product, tourism-based activities are also among the principal causes of land reclamation and degradation (MA, 2005b). Hence it is necessary to find sustainable management plans to balance economic growth, ecosystem health and integrity, and providing ecosystem services and societal goods. The ES approach can be a powerful tool to coastal and estuarine management plans, by providing scientifically-grounded information and simple and straightforward outputs capable of engaging different stakeholders and helping decision-makers. The use of an ES approach in decision-making (MA, 2005b; de Groot et al., 2010; Barbier et al., 2011; Beck et al., 2011), by means of participatory mapping (Klain and Chan, 2012; van Riper et al., 2012; Hernández-Morcillo et al., 2013; Arkema et al., 2015), or through modelling and visualization tools such as InVEST, can improve decision-making to better tackle sustainability problems. These methods are a valuable tool to help decision-makers better visualise areas and valuable hotspots. They also indicate where to prioritise actions to protect a given ecosystem or on the other hand, where to encourage development of

structures with a lesser impact on ecosystems. By providing synergies between the protection offered by local reserves together with the recreation potential that these areas have in the public opinion could be the way forward to a better integrated coastal management (Chan and Ruckelshaus, 2010).

Our first assessment of the recreation patterns in the NW of Portugal is an important step to assist in better ecosystem management decisions and land use planning and it also considers the non-monetary values of ecosystems (Tenerelli et al., 2016). This is particularly important in the case of the NW of Portugal, where different portions of the territory are under national or EU based regulation. Municipalities and other entities could also take these new approaches for data gathering and analysis of visit data, an approach more cost-effective than traditional methods (Tenerelli et al., 2016). The visitor and temporal pattern data could also be valuable, for example, when evaluating transport or assessing visit frequency and management of different areas of the territory and ecosystems according to visitor patterns (Wood et al., 2013; Sessions et al., 2016). Moreover, and more importantly, against a background of the EU 'Blue Growth Agenda' (EC, 2014), the marine, coastal and estuarine ES approach could be fundamental to support and guide management decisions in the maritime sectors and support EU Directives such as the Maritime Spatial Planning Directive (Directive 2014/89/EU).

The approach described here could be taken further using other types of social media data sources (e.g. Pinterest) to infer information about visit data. By linking the available data, especially monetary information on Nature-based recreation, scientists could effectively attribute 'monetary value' on these unequivocal intrinsic values of Cultural Ecosystem Services.

5. Conclusion

The present study showed that coastal and estuarine ecosystems were important providers of recreation and tourism of the NW Portugal, particularly the Douro estuary that was highly associated with the tourism increase in the city of Porto. Although providing valuable information for our indicators of visitor preferences, the results showed that the InVEST visitor outputs might have underestimated the visitor numbers after 2011. In contrast, socioeconomic tourism indicators showed a continuous increase of visitors in the study area. This emphasises the importance of combining different methodologies in ES assessments but also emphasises the need to continue to seek broader and holistic approaches when quantifying marine cultural ES. We have shown how the same type of ecosystems, such as estuaries, contributed differently for recreation, highlighting the importance of local ES assessments instead of the continental or global approaches more commonly used in this type of study. As such, the local approach provides local up-to-date information, ES spatial awareness and the societal need for better-grounded judgment and decision-making.

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